

WINDISCHGARSTEN, Austria

WMO: 113550

Lat: 47.720N

Lon: 14.326E

Elev: 598

StdP: 94.35

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -14.1	(c) -11.5	(d) -15.9	(e) 1.0	(f) -12.0	(g) -13.4	(h) 1.3	(i) -10.5	(j) 8.2	(k) 4.1	(l) 7.1	(m) 3.8	(n) 0.7	(o) 80	(p) 0.501

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	30.4	20.2	28.4	19.3	26.6	18.3	20.9	28.5	20.1	26.8	19.1	25.1	2.5	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	14.5	23.4	17.7	13.6	22.5	16.9	13.0	21.6	63.2	28.6	60.1	27.0	56.9	25.2	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.1	5.7	4.6	-16.7	33.7	4.1	1.7	-19.6	34.9	-22.0	35.9	-24.3	36.8	-27.2	38.0
			-16.9	22.5	4.0	1.0	-19.8	23.1	-22.1	23.7	-24.4	24.3	-27.3	25.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	-2.1	0.2	4.0	8.9	13.4	16.9	18.3	18.1	13.8	9.8	4.2	-0.4
	DBStd	8.20	4.44	4.44	4.03	4.33	3.86	3.88	3.44	3.38	3.51	3.83	4.62	4.28
	HDD10.0	1489	376	273	188	70	13	2	0	1	8	52	183	323
	HDD18.3	3601	634	507	444	284	159	72	43	46	141	266	426	581
	CDD10.0	1049	0	0	2	37	118	208	259	251	122	45	8	0
	CDD18.3	119	0	0	0	1	5	28	43	38	4	0	0	0
	CDH23.3	1378	0	0	0	14	97	334	459	410	59	4	0	0
	CDH26.7	392	0	0	0	1	16	100	144	123	8	0	0	0
(14) Wind	WSAvg	1.4	1.3	1.5	1.6	1.8	1.6	1.4	1.4	1.3	1.2	1.4	1.4	1.2
(15) Precipitation	PrecAvg	1238	74	70	93	73	116	146	166	148	113	84	79	77
	PrecMax	1441	181	183	215	135	222	233	335	273	234	173	207	215
	PrecMin	922	5	18	31	15	32	73	57	64	44	23	1	20
	PrecStd	129	49	41	53	29	47	41	71	49	47	36	43	43
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	13.7	18.8	25.2	28.8	32.2	33.0	32.5	27.7	23.7	19.4	13.7
		MCWB	7.1	7.1	10.4	14.8	18.0	20.8	21.4	20.8	18.4	15.4	12.0	8.2
	2%	DB	8.2	10.8	16.3	22.3	26.1	29.4	30.2	30.0	25.0	20.8	16.6	10.4
		MCWB	4.6	5.7	9.4	13.1	16.6	19.7	20.3	20.1	17.2	14.2	10.7	6.0
	5%	DB	5.4	8.8	13.6	19.8	23.8	27.3	28.0	27.6	22.9	18.7	14.2	7.4
		MCWB	3.4	4.6	7.9	11.9	15.3	18.5	19.3	19.2	16.2	12.9	9.4	5.0
	10%	DB	3.4	6.5	11.0	17.3	21.5	24.9	25.9	25.5	20.8	16.6	11.5	5.0
		MCWB	2.2	3.5	6.7	10.7	14.2	17.5	18.5	18.3	15.3	11.8	8.0	3.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.6	7.9	11.1	15.1	18.9	21.8	22.2	21.9	19.5	16.0	12.5	8.6
		MCDB	10.9	12.3	17.2	24.2	27.1	29.9	30.8	29.4	25.4	21.6	18.3	13.1
	2%	WB	5.0	6.2	9.7	13.6	17.3	20.4	21.0	20.7	18.0	14.6	11.0	6.8
		MCDB	7.3	10.1	15.2	21.4	24.2	27.7	28.5	28.0	23.4	19.5	15.9	9.5
	5%	WB	3.6	4.9	8.4	12.3	16.0	19.2	20.1	19.8	17.0	13.6	9.7	5.1
		MCDB	5.2	8.1	12.7	18.8	22.1	25.6	26.7	26.3	21.5	17.5	13.6	7.3
	10%	WB	2.3	3.7	7.1	11.1	14.8	18.1	19.0	18.8	15.8	12.6	8.3	3.6
		MCDB	3.3	6.2	10.3	16.5	20.4	23.7	24.6	24.4	19.8	15.9	11.1	5.1
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	8.6	9.7	10.9	10.7	10.6	10.9	11.2	10.3	10.4	7.8	6.8
		MCDBR	9.4	13.0	15.9	16.5	16.2	16.0	15.8	16.0	14.9	14.7	11.7	10.1
	5% WB	MCWBR	6.7	8.3	9.2	8.3	7.5	6.9	6.6	6.9	7.7	8.4	6.9	6.9
		MCWBR	8.3	11.2	14.4	15.2	14.4	14.5	14.2	14.6	13.6	13.1	10.9	9.4
(40) Clear-Sky Solar Irradiance	taub		0.283	0.312	0.352	0.390	0.400	0.405	0.410	0.402	0.373	0.342	0.314	0.284
		taud	2.395	2.337	2.301	2.271	2.287	2.309	2.299	2.337	2.393	2.459	2.483	2.452
	Ebn at Noon		802	847	862	861	864	859	849	839	832	806	760	759
		Edh at Noon	68	89	109	124	127	125	125	115	99	80	63	58
(44) All-Sky Solar Radiation	RadAvg		1.16	2.05	3.08	4.15	4.80	5.25	5.22	4.66	3.37	2.28	1.25	0.95
	RadStd		0.14	0.26	0.35	0.46	0.55	0.44	0.49	0.53	0.43	0.23	0.17	0.13

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (104 neighbors)		+0.52	N/A	N/A	+0.78	+0.45	+0.40	-110	-142	+81

Nomenclature: See separate page